

Дано:

$$\omega O_1 \cap \omega O_2 = A, B$$

$$(CA) \cap \omega O_2 = D$$

$$(BC) \cap \omega O_2 = E$$

Дока-те:

$$\triangle CBD \sim \triangle O_1 A O_2$$

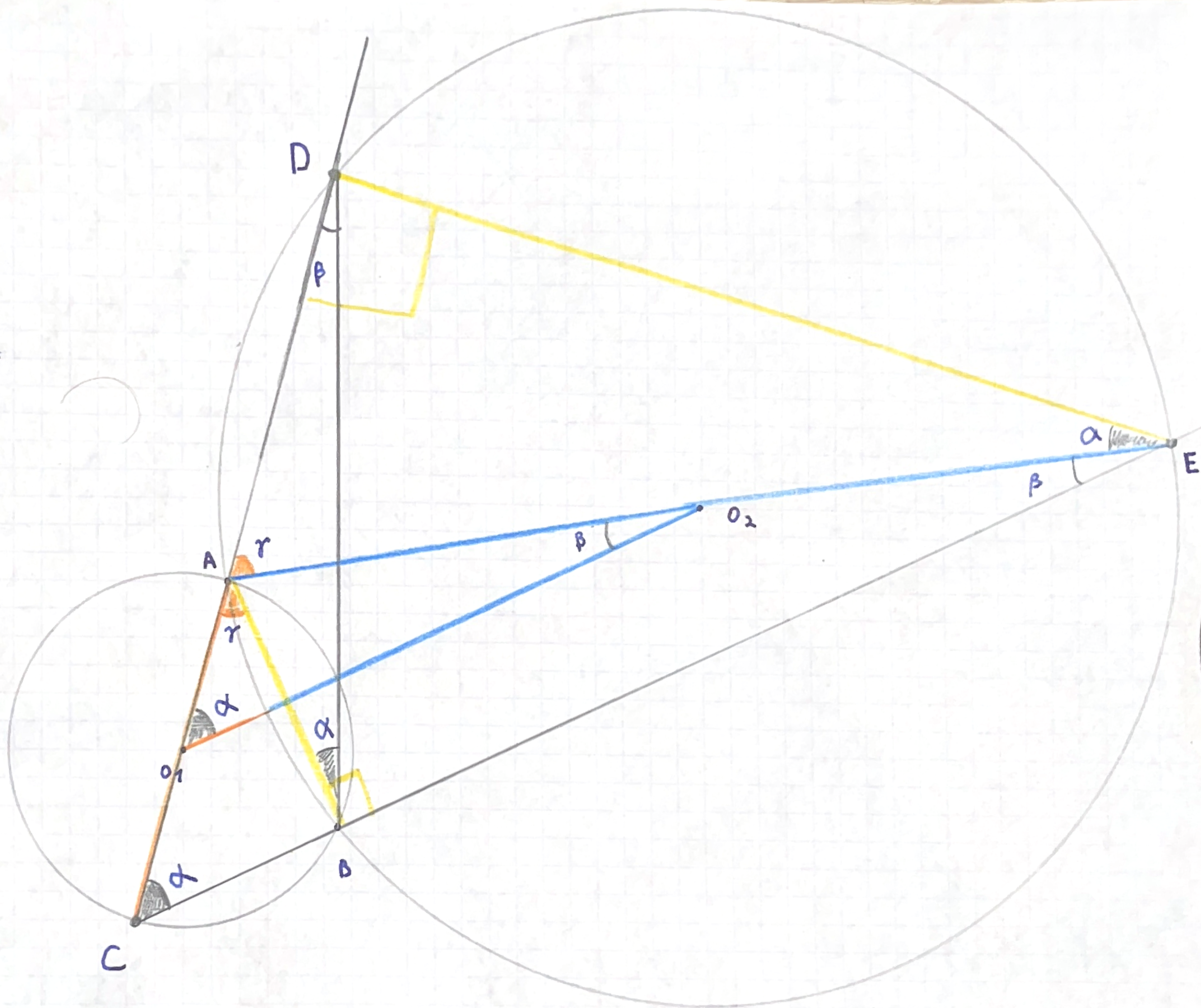
Найти:

$$AD = ?$$

$$\angle PAE = \angle BAC (*)$$

$$AB = 3 (*)$$

$$R_2 = 3 R_1 (*)$$



Демонстрация :

$$\begin{array}{l} AO_1 = O_1 C \\ AO_2 = O_2 E \end{array} \left| \Rightarrow O_1 O_2 - \text{ср. линия} \Rightarrow O_1 O_2 \parallel CE \Rightarrow \angle AO_1 O_2 = \angle ACE \right.$$
$$\angle AO_2 O_1 = \angle AEC$$

$$\angle AEC = \angle APB \text{ (определение на } \angle A B)$$

$$\begin{array}{l} \angle AEC = \angle APB \\ \angle AO_1 O_2 = \angle ACE \end{array} \left| \Rightarrow \triangle CBD \sim \triangle O_1 A O_2 \quad \text{Доказано} \right.$$

Проведем DE : $\angle ADE = 90^\circ$; $\angle ABE = 90^\circ$ (определение на диаметр)

$$\begin{array}{l} \angle CAB = \angle DAE \\ \angle ADE = \angle CBA \end{array} \left| \Rightarrow \triangle CAB \sim \triangle AED \Rightarrow \angle DEA = \alpha \Rightarrow \angle ABD = \alpha \text{ (на одной дуге)} \right.$$

$$\sin \alpha = \frac{3}{2R_1}$$
$$\sin \beta = \frac{3}{6R_1} = \frac{1}{2R_1}$$

По теореме синусов в $\triangle ADB$:

$$\frac{AD}{\sin \alpha} = \frac{AB}{\sin \beta} \Rightarrow AD = \frac{AB \cdot \sin \alpha}{\sin \beta} = 3 \cdot \left(\frac{3}{2R_1} : \frac{1}{2R_1} \right) = 9$$